Revisiting MR^3 for the Bidiagonal SVD

Recent Progress and Open Directions in Matrix Computations

Ryan Schneider | UC Berkeley
September 23rd 2025

Our starting point

The symmetric eigenvalue problem (SEP)

$$A = QDQ^T$$

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$n \times n$ (real) symmetric matrix

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Computing the SVD of an arbitrary (real) matrix G reduces to diagonalizing the symmetric matrices G^TG – e.g., $G = U\Sigma V^T \iff G^TG = V\Sigma^2V^T$.

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Bonus points for parallelism/the ability to compute a subset of eigenpairs.

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    - Cons: explicit (and expensive) re-orthogonalization may be necessary.

The challenge of orthogonal eigenvectors

Suppose $\lambda_1, \lambda_2, \ldots \lambda_n$ are the eigenvalues of T, corresponding to eigenvectors $v_1, v_2, \ldots, v_n$. If (λ, v) is an approximate eigenpair and θ_i is the angle between v and v_i then

$$\sin(\theta_i) \leq \frac{||(T - \lambda I)v||_2}{\min_{j \neq i} |\lambda - \lambda_j|}$$

To guarantee orthogonality we need (1) a small relative eigenvalue gap and (2) tighter residual error $||(T - \lambda I)v||_2 = O(\varepsilon n |\lambda_i|)$.

Enter MR^3

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$$(T - \hat{\lambda} I)x = \delta_k e_k$$

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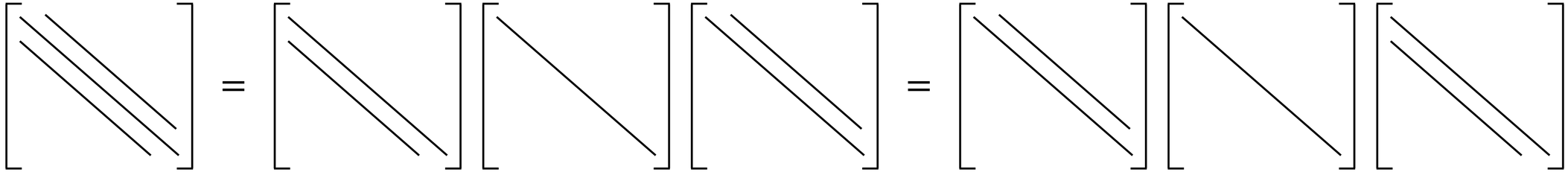

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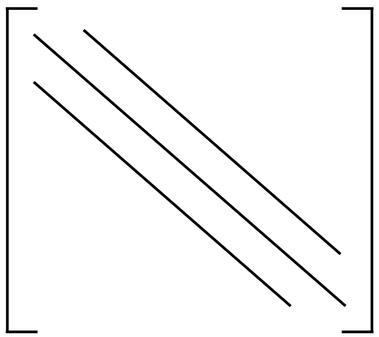

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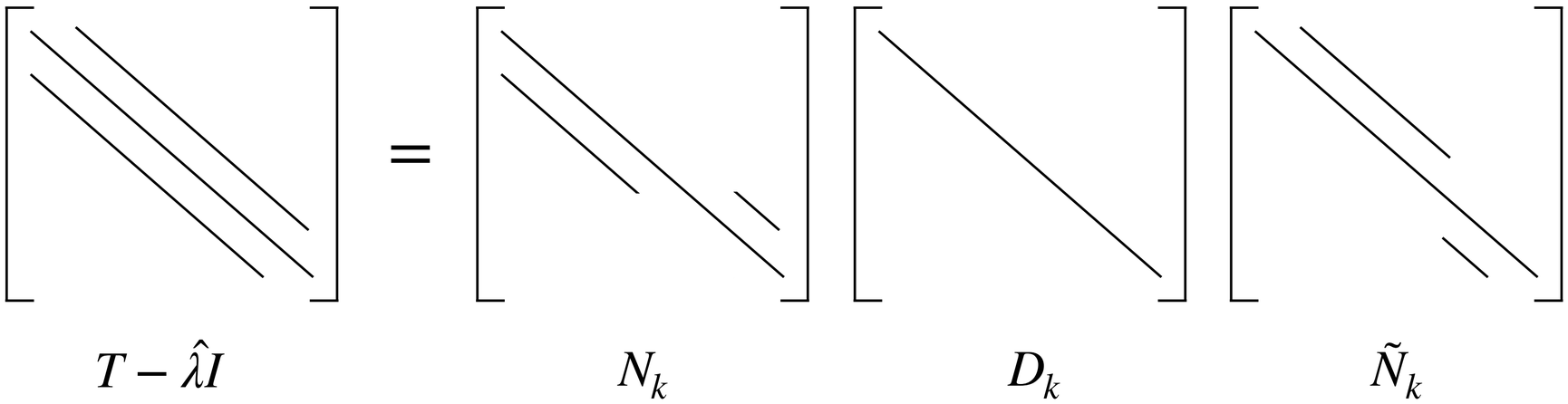

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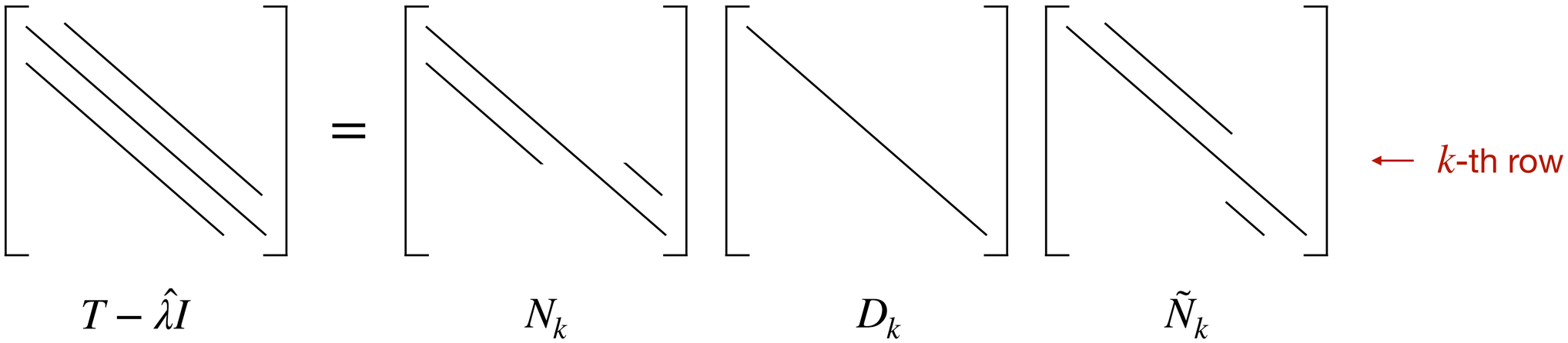

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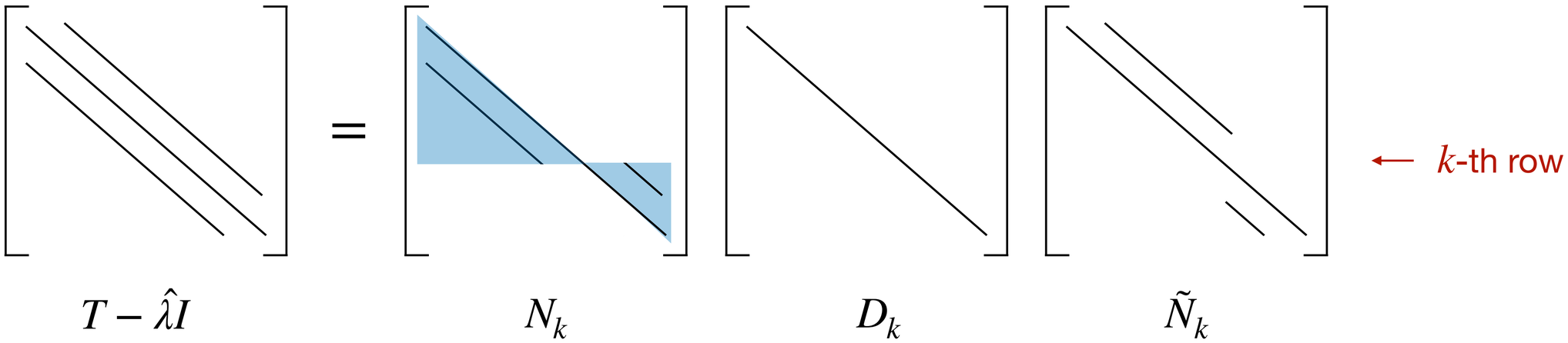

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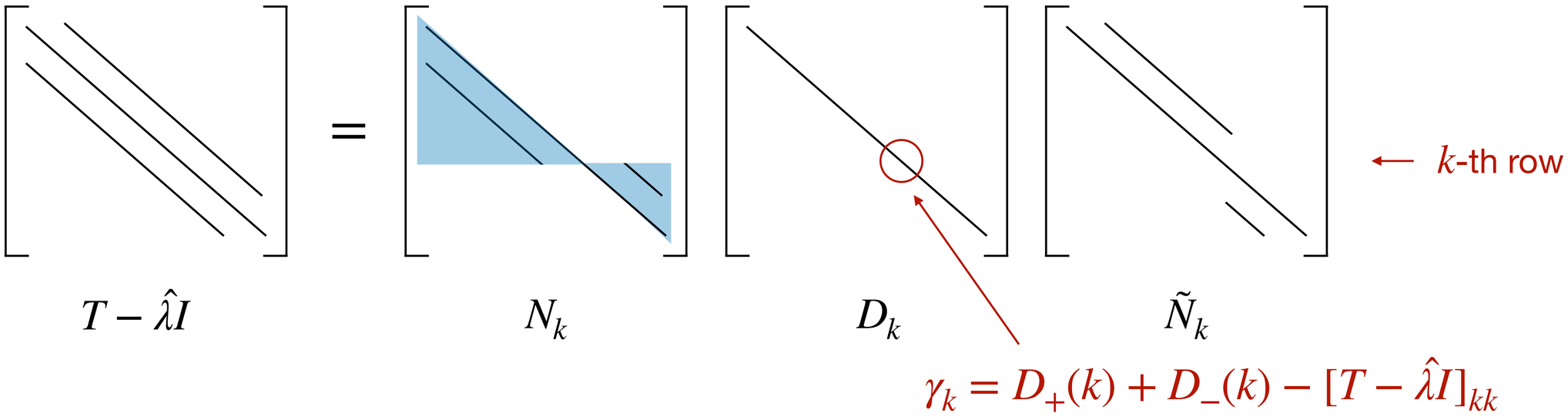

$$\gamma_k = D_+(k) + D_-(k) - [T - \hat{\lambda} I]_{kk}$$

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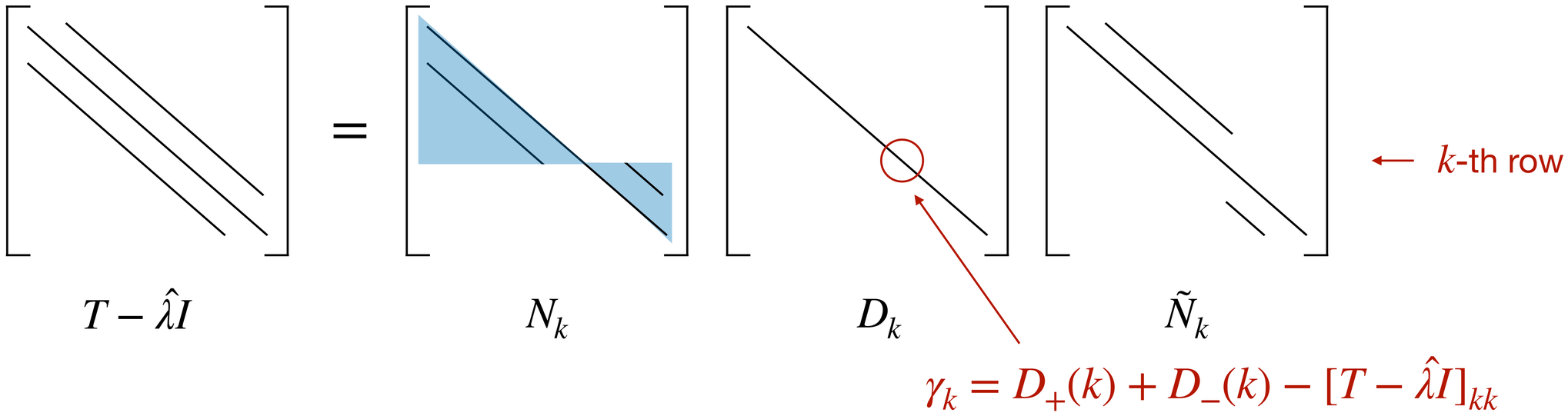

Choose k such that $|\gamma_k|$ is minimized and set $\delta_k = \gamma_k$.

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$$\tilde{x}_i = (1 + \epsilon_i)x_i \implies \frac{|\tilde{\lambda}_j - \lambda_j|}{|\lambda_j|} = O\left(\sum_i \epsilon_i\right), \quad \sin(\theta_j) = O\left(\frac{\sum_i \epsilon_i}{\text{relgap}(\lambda_j)}\right)$$

angle between v_j and $\tilde{v}_j$

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The final ingredient: To obtain a relative eigenvalue gap for any eigenvalue, simply shift T!

MR^3: The Full Algorithm

Input: Symmetric tridiagonal T

1. Shift and factor: $T + \mu I = LDL^T$
2. Compute the eigenvalues of LDL^T
3. For eigenvalues that are *relatively* isolated: apply one-step inverse-iteration via twisted factorizations of LDL^T
4. For clustered eigenvalues: shift, compute a new RRR, and return to 2.

Output: Numerically orthogonal eigenvectors!

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left/right singular vectors

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$$G = (Q_L U) \Sigma (Q_R V)^T$$

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Goal: Given B, find triples (σ_i, u_i, v_i) such that

- $||Bv_i - \sigma_i u_i||_2 = O(\varepsilon n ||B||_2)$
- $|v_i^T v_j| = |u_i^T u_j| = O(\varepsilon n), \ i \neq j$

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Bidiagonal SVD $\longleftrightarrow$ Tridiagonal SEP

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3. $\begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix} = Q \begin{pmatrix} -\Sigma & 0 \\ 0 & \Sigma \end{pmatrix} Q^T$ for $Q = \frac{1}{\sqrt{2}} \begin{pmatrix} U & U \\ -V & V \end{pmatrix}$

Bidiagonal SVD $\longleftrightarrow$ Tridiagonal SEP

If $B = U\Sigma V^T$ then…

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Applying a perfect shuffle permutation on both sides yields the Golub-Kahan matrix T_{GK}

The pitfalls of using MR^3 as a black box

Option 1

Option 2

The pitfalls of using MR^3 as a black box

Option 1

1. Apply MR^3 to $B^T B$ (for V)
2. Apply MR^3 to BB^T (for U)

Option 2

The pitfalls of using MR^3 as a black box

Option 1

1. Apply MR^3 to $B^T B$ (for V)
2. Apply MR^3 to BB^T (for U)

Option 2

1. Form the Golub-Kahan matrix T_{GK}
2. Apply MR^3 to T_{GK}
3. Extract U and V

The pitfalls of using MR^3 as a black box

Option 1

1. Apply MR^3 to $B^T B$ (for V)
2. Apply MR^3 to BB^T (for U)

❌ $||Bv_i - \sigma_i u_i||_2 = O(\varepsilon n ||B||_2)$

✅ $|v_i^T v_j| = |u_i^T u_j| = O(\varepsilon n), \ i \neq j$

Option 2

1. Form the Golub-Kahan matrix T_{GK}
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The pitfalls of using MR^3 as a black box

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Enter MR^3

In essence, the Multiple Relatively Robust Representations (MR^3 or MRRR) algorithm is a version of bisection and inverse iteration that can compute k eigenpairs in only $O(kn)$ operations, *without* explicitly orthogonalizing.

History:

- 1997: A New $O(n^2)$ Algorithm for the Symmetric Tridiagonal Eigenvalue/Eigenvector Problem (Dhillon, PhD Thesis)
- 2000: Relatively Robust Representations of Symmetric Tridiagonals (Parlett & Dhillon, LAA)
- 2004: Orthogonal Eigenvectors and Relative Gaps (Dhillon & Parlett, SIMAX)
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  see [Marques, Demmel, & Vasconcelos 2020]

Failure Example

[Marques, Demmel, & Vasconcelos]

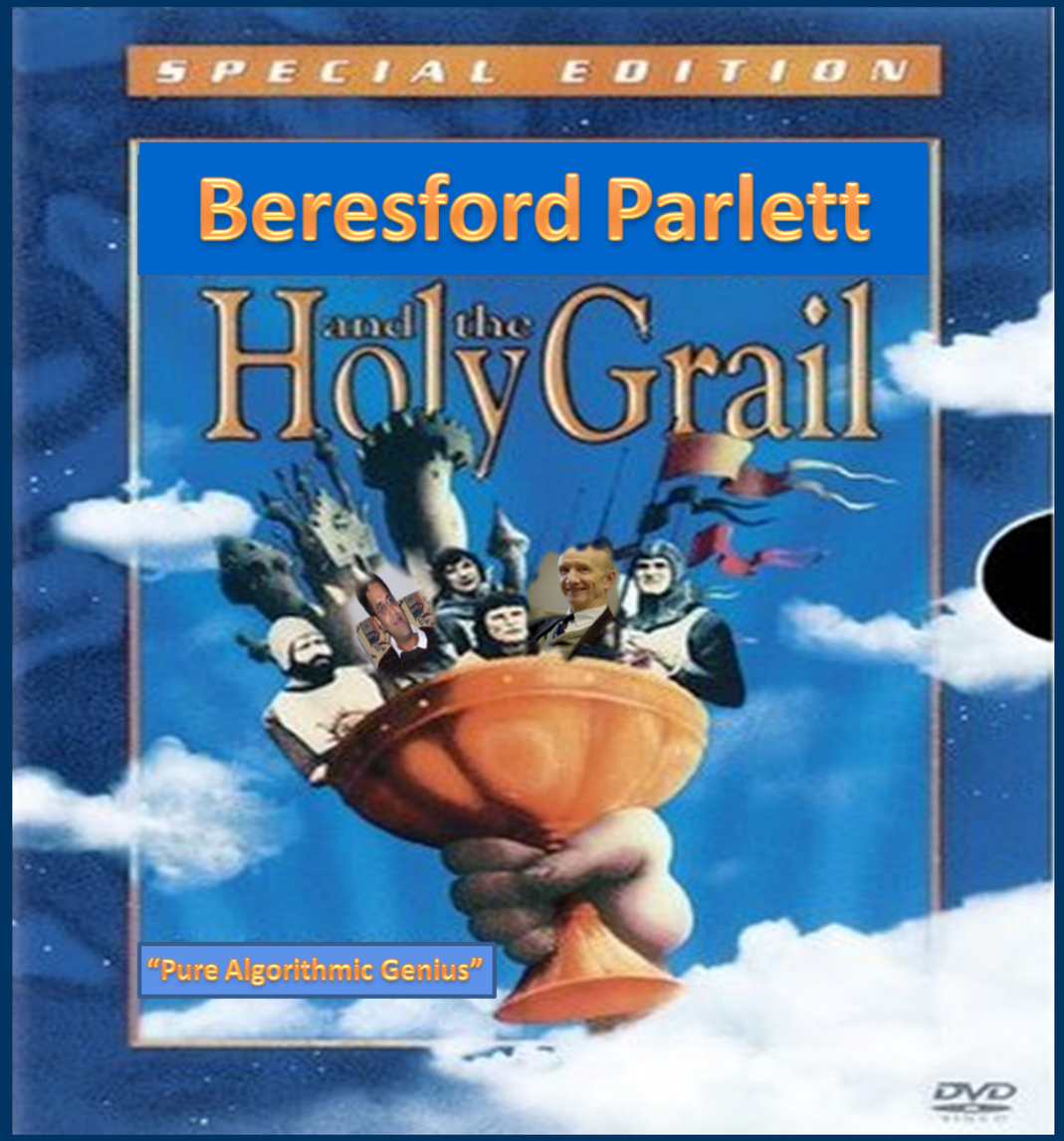
T is a 10×10 tridiagonal with eigenvalues

$$\lambda_j = \left(\frac{1}{\sqrt{\epsilon}} \right)^{-\frac{j-1}{9}}$$

Computed eigenvectors V satisfy only $||I - V^T V||_2/(n\epsilon) = 3.95 \times 10^4$.

Open questions

1. Can we finally get MR^3 working in this setting?
    - Debug existing implementations
    - Revisit the theory
    - Randomize?
2. Other potential applications of MR^3:
    - Preconditioned Jacobi's method for the bidiagonal SVD?

Thank you!

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